

EduVision_DV QA Checklist

Higher Education Performance Analytics | Final Testing & Validation

Test ID	Area	Test Case	Expected Result	Actual/Evidence	Status	Notes
QA-001	Data	dim_university university_id uniqueness	No duplicate university IDs	0 duplicate IDs; 4,708 unique IDs	PASS	Verified from final Excel
QA-002	Data	dim_country country_id uniqueness	No duplicate country IDs	0 duplicate IDs; 136 unique IDs	PASS	Verified from final Excel
QA-003	Data	Fact table composite key	No duplicate university_id + year + source	0 duplicates in KPI/performance/research/student facts	PASS	Composite key used because one university has multiple years/sources
QA-004	Data	University referential integrity	Every fact university_id exists in dim_university	0 invalid university IDs in fact tables	PASS	Verified
QA-005	Data	Country referential integrity	Every non-null country_id exists in dim_country	0 invalid country IDs	PASS	Verified
QA-006	Data	Missing values	Missing values are identified and documented	Missingness remains in several source indicators	PASS WITH NOTE	Missing does not mean zero; document as source limitation
QA-007	KPI	Global Rank KPI	Matches defined aggregation	Dashboard shows 1.00; MIN(global_rank)=1	PASS	Independent recomputation
QA-008	KPI	Overall Score KPI	Matches defined aggregation	52.50 vs AVG=52.5008	PASS	Rounded display
QA-009	KPI	Academic Reputation KPI	Matches defined aggregation	20.29 vs AVG=20.2867	PASS	Rounded display
QA-010	KPI	Research Impact KPI	Matches defined aggregation	37.22 vs AVG=37.2192	PASS	Rounded display
QA-011	KPI	Faculty/Student Ratio KPI	Matches defined aggregation	19.07 vs AVG=19.0717	PASS	Rounded display
QA-012	KPI	International Student % KPI	Matches defined aggregation	9.95% vs AVG=9.9487%	PASS	True percentage field
QA-013	KPI	Research Score KPI	Matches defined aggregation	23.38 vs AVG=23.3824	PASS	Independent recomputation
QA-014	KPI	Citation Score KPI	Matches defined aggregation	42.98 vs AVG=42.9805	PASS	Independent recomputation
QA-015	KPI	Research Productivity KPI	Matches defined aggregation	50.03 vs AVG=50.0282	PASS	Derived KPI already present in final model
QA-016	KPI	Total Students KPI	Matches sum of total_students	92.41M; Excel sum=92,405,187	PASS	Rounded to 2 decimals
QA-017	Country KPI	Total Countries	Matches COUNTD country_id	136	PASS	Independent recomputation
QA-018	Country KPI	Education Countries	Matches fact_country_education count	119	PASS	Independent recomputation
QA-019	Country KPI	Average Literacy	Matches average literacy field	87.97 vs AVG=87.9667	PASS	Rounded display
QA-020	Country KPI	Average Tertiary Enrollment	Matches average tertiary enrollment	44.33 vs AVG=44.3255	PASS	Rounded display
QA-021	Dashboard	Four dashboards present	All four required dashboards available	University Overview, Research Analytics, Student Analytics, Country Comparison	PASS	Screenshot evidence
QA-022	Dashboard	Navigation buttons	Each dashboard can be opened from navigation	Buttons visible on left panel	PASS	Verified manually in Tableau by project owner.
QA-023	Dashboard	Year filter	Changing year updates relevant views	Year filter visible on dashboard screenshots	PASS	Verified manually in Tableau by project owner.
QA-024	Dashboard	University/Country filters	Filter changes update views	Filters visible where applicable	PASS	Verified manually in Tableau by project owner.

Test ID	Area	Test Case	Expected Result	Actual/Evidence	Status	Notes
QA-025	Dashboard	Dashboard actions	Selections carry to linked dashboards where configured	Integration configured	MANUAL	Test Oxford/one country path
QA-026	Dashboard	No broken visuals	Charts render without errors	All supplied screenshots render normally	PASS	No broken-object errors visible
QA-027	Dashboard	Visual quality	No major layout failures	Dashboards readable and consistently formatted	PASS WITH NOTE	Some scrollbars/null categories remain
QA-028	Documentation	KPI definitions	Every KPI has source/formula/unit/interpretation	Included in final documentation	PASS	Submission pack
QA-029	Documentation	Limitations	Known missingness and source limitations documented	Included in final documentation	PASS	Submission pack
QA-030	Delivery	Workbook	Final .twbx saved and opens	EduVision_DV.twbx required	MANUAL	Attach actual workbook

QA Test Summary

The four dashboard interaction tests were manually checked by the project owner and are confirmed working. The remaining checklist items document data, KPI, dashboard, and delivery validation.

Metric	Value
Total test cases	30
PASS	26
PASS WITH NOTE	2
MANUAL final click-through	2
KPI calculations independently recomputed	Yes – displayed KPI values
KPI accuracy claim	Do not claim >95% unless manual sample testing is completed
Known data limitation	Missing values/region categories exist and are documented

Final manual interaction verification: PASS – navigation, Year filter, University/Country filters, and dashboard linking were checked successfully.